Submodule 40.04: Landmark CNN Architectures: From LeNet-5 to ResNet

1. The Timeline of Deep Vision Breakthroughs

The history of Computer Vision is defined by milestone architectures that introduced foundational design patterns used across modern deep learning.

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1998            2012            2014            2014            2015
LeNet-5 ------> AlexNet ------> VGG-16 -------> GoogLeNet -----> ResNet
(Yann LeCun)    (Krizhevsky)    (Simonyan)      (Inception)     (Kaiming He)
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2. Deep Dive into the 5 Landmark Architectures

1. LeNet-5 (1998, Yann LeCun et al.)

- **Pioneering Achievement**: First commercially successful convolutional neural network, deployed by US banks to read handwritten checks on the MNIST dataset.
- **Key Structure**: 2 Conv layers with 5×5 filters + Average Pooling + 2 Fully Connected layers.
- **Activation**: Sigmoid / Tanh functions (predating ReLU).

2. AlexNet (2012, Alex Krizhevsky, Ilya Sutskever, Geoffrey Hinton)

- **The ImageNet Revolution**: Won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC 2012) by an unprecedented margin, cutting the top-5 error rate from 26% to 15.3%.
- **Key Innovations**:
    1. First major model trained entirely on parallel **NVIDIA GPUs** (using custom CUDA kernels).
    2. Adopted **ReLU** instead of Sigmoid, accelerating training convergence by $6\times$.
    3. Introduced **Dropout ($p = 0.5$)** to combat catastrophic overfitting in dense layers.
    4. Data Augmentation (random crops, horizontal flips, color jittering).

3. VGG-16 / VGG-19 (2014, Visual Geometry Group, Oxford)

- **Core Philosophy**: Extreme simplicity and architectural uniformity.
- **The 3×3 Factorization Principle**:
    - Replaced large kernels (5×5 and 7×7) with stacks of small 3×3 kernels with stride 1 and padding 1.
    - **Mathematical Proof of Advantage**: Two stacked 3×3 convolutions have the exact same effective receptive field as a single 5×5 convolution:

$$\text{Receptive Field} = 1 + (3 - 1) + (3 - 1) = 5$$

        - Parameters for two 3×3 filters: $2 \times (3 \times 3 \times C^2) = \mathbf{18C^2}$
        - Parameters for one 5×5 filter: $1 \times (5 \times 5 \times C^2) = \mathbf{25C^2}$
        - **Result**: 28% fewer parameters AND two non-linear ReLU activations instead of one!

4. GoogLeNet / Inception-v1 (2014, Google)

- **Key Innovation**: The **Inception Module**—processing features at multiple spatial scales simultaneously.
- Rather than choosing between $1 \times 1, 3 \times 3$, or 5×5 filters, the Inception module runs all of them in parallel and concatenates their output feature maps!
- Introduced 1×1 **Bottleneck Convolutions** to reduce channel dimensions before costly 3×3 and 5×5 convolutions, keeping parameter count to just 5 Million (vs AlexNet's 60M).

5. ResNet (2015, Kaiming He et al., Microsoft Research)

- **The Problem of Network Degradation**: As neural networks grow deeper (e.g. 50 to 100+ layers), accuracy saturates and degrades rapidly. This is NOT caused by overfitting (training error also degrades!), but by **vanishing gradients** and extreme optimization difficulty.
- **The Solution: Residual Skip Connections**: Instead of forcing stacked layers to fit a desired underlying mapping $\mathcal{H}(\mathbf{x})$, ResNet forces them to fit a **residual mapping** $\mathcal{F}(\mathbf{x}) = \mathcal{H}(\mathbf{x}) - \mathbf{x}$:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x}$$

    - $\mathbf{x}$ is the **identity shortcut connection** (skipping one or more layers).

- During backpropagation:

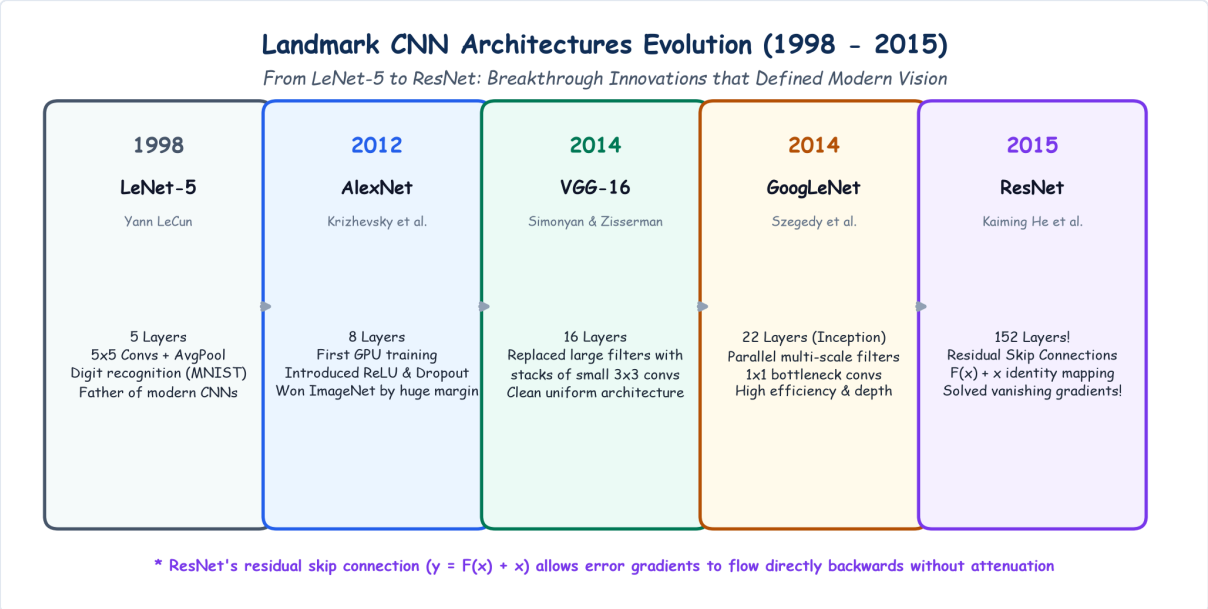
$$\frac{\partial L}{\partial \mathbf{x}} = \frac{\partial L}{\partial \mathbf{y}} \frac{\partial \mathcal{F}}{\partial \mathbf{x}} + \frac{\partial \mathbf{L}}{\partial \mathbf{y}}$$

- The second term ($+\frac{\partial L}{\partial \mathbf{y}}$) acts as an unobstructed **gradient highway**! Gradients flow directly backward without decaying, enabling networks with **152+ layers** to be trained stably.

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3. Hand-Drawn Educational Diagram: Architectural Evolution

The following notebook-style diagram traces the evolutionary breakthroughs across all 5 landmark architectures.



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4. Summary Comparison Table

Architecture	Year	Total Layers	Key Architectural Breakthrough	Top-5 ImageNet Error
LeNet-5	1998	5	First practical CNN pipeline (Conv + Pool)	N/A (MNIST)
AlexNet	2012	8	GPU training, ReLU activation, Dropout regularization	15.3%
VGG-16	2014	16	Factorization into homogeneous stacks of 3×3	7.3%

Architecture	Year	Total Layers	Key Architectural Breakthrough	Top-5 ImageNet Error
			convs	
GoogLeNet	2014	22	Parallel Inception modules + 1×1 bottleneck convs	6.7%
ResNet-152	2015	152	Residual skip connections ($F(x) + x$)	**3.57%** (Surpassed Human Vision!)